An Hourly AI Risk Index and Its Relationship with AI-Company Returns

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Abstract

This report implements the AI-risk branch of a broader project on how AI risk and AI development may relate to AI-company returns. It converts 1,882 X News story clusters into a timestamped, ex-ante-eligible hourly text measure by labeling 1,882 headlines and 13,284 summary clauses with DeepSeek, scoring both concrete and generic risk language, applying strictly prior-only RMS scaling, and aggregating documents by hour. The observed X News sample is aligned to six New York trading windows per day for 20 AI-exposed stocks and QQQ. Panel and equal-weight Granger-style regressions find no statistically reliable next-hour return relationship at one to three risk lags. Category tests are likewise null after multiple-testing control. A high-risk long-QQQ/short-AI strategy loses 0.38% before costs and 2.45% after a 5 bp one-way turnover charge. These results reject neither the broader existence nor the long-horizon relevance of AI risk; they show that this particular short-sample, X-based hourly factor is not yet a robust predictor or tradable signal. The proposed AI-development factor remains unimplemented and is therefore not tested.

Keywords: artificial intelligence risk; news analytics; large language models; intraday returns; Granger predictability; text-based factor

1 Introduction

The project begins with two directional hypotheses: news about AI-related harm may depress the returns of AI-exposed firms, whereas news about AI development may support them. This framing follows a large literature showing that media language can contain or amplify information relevant to financial prices and trading activity (Tetlock, 2007; Sacerdote, Sehgal, and Cook, 2020). It also motivates a predictive design rather than a claim that news mechanically causes prices.

The central contribution is a transparent measurement chain from text evidence to an hourly index: theory defines five harms; an LLM classifies headlines and already-split clauses; each output retains a reason and verbatim evidence; document-level scores are scaled only with information available before the scoring hour; and the resulting series is lagged before it enters a return model. The analysis therefore makes the measurement assumptions visible rather than treating a generic sentiment score as AI risk.

1.1 Scope and interpretation

The completed empirical work concerns AI risk only. The project brief proposes an AI-development hypothesis but supplies no final operational definition, labels, index, or empirical test for it. This report therefore does not manufacture a development result. Likewise, all reported coefficients are short-horizon predictive associations in the available sample, not structural causal effects.

2 News Data and LLM Annotation

2.1 Corpus construction

Two news corpora were collected with the same 33 AI-related keyword queries. X News supplies generated story clusters, short summaries, linked post evidence, and a cluster updated_at timestamp. It does not supply an independent publisher article for every record. Guardian supplies full article HTML. Only the X corpus has completed the full labeling, index, regression, and backtest pipeline; Guardian is retained for later validation and extension.

Source	Collection window	Documents	Sentences	Clauses	Role in this report
X News	Sep 2-Oct 2, 2026 (index time)	1,882 clusters	8,126	15,281; 13,284 body targets	Full empirical pipeline
Guardian	Sep 1-Sep 30, 2026	303 articles	24,873	44,224; 43,750 body clauses	Collected and split only

Notes: All 33 X queries reached the 100-result cap. X returned 14,169 linked post IDs, but only 2,223 evidence posts were retrieved and no reaction-post sample was collected. The X timestamp is the cluster update time, not necessarily the publisher release time.

2.2 Annotation protocol

DeepSeek deepseek-v4-pro labeled each headline and each pre-existing body clause as a separate target under prompt version topic_risk_stance_v11_2026-10-02 (temperature 0, JSON output, thinking disabled). The model could use the original sentence only to resolve reference, negation, condition, or quotation; it could not transfer independent risk content from neighboring clauses or the headline. Program-side checks enforced allowed values, evidence copied verbatim from the inputs, conditional field dependencies, retries, and resumable persistence.

Topic value	Meaning	Risk type	Direction
0	No supported link to the five in-scope AI harms	null	null / not applicable
0.2	Generic AI-risk language without evidence for a specific category	No category assigned	+1, 0, or -1 stance
1	Concrete reference to at least one in-scope harm	One or more of five categories	+1 affirming; 0 neutral/mixed; -1 denying or mitigating

The five concrete categories are loss of human control, malicious use, job displacement, discrimination, and privacy violation. A review flag is separate from the score: it identifies ambiguity or mixed readings and is never treated as a zero by default.

2.3 Label distribution

Unit	Total	Topic = 0	Topic = 0.2	Topic = 1	Review
Headlines	1,882	1,681	124	77	66
Body clauses	13,284	12,437	383	464	728
Combined	15,166	14,118	507	541	794

Only 6.91% of units are broadly relevant (0.2 or 1). Among those 1,048 units, 708 affirm or warn about risk, 202 are neutral/mixed/unclear, and 138 deny or mitigate risk. There are 794 review flags, but only three flagged clauses contribute a nonzero raw score.

3 Hourly AI-Risk Index

3.1 From annotated units to documents

For unit c in document i, the signed raw contribution is topic relevance times risk direction. A generic risk unit therefore contributes at most 0.2 in magnitude, while a concrete category unit contributes at most 1. The headline contribution is H; body contributions are summed into B. Category components include only units explicitly assigned to that category.

$$u_{ic} = relevance_{ic} \times direction_{ic}; \quad H_i = u_{i,headline}; \quad B_i = Sum_{c \text{ in body}(i)}\, u_{ic}$$

Headline and body scales can differ mechanically, so each is divided by a root-mean-square scale computed only from earlier retained documents. The procedure does not subtract a mean. Scaling begins only after 100 earlier documents and five earlier nonzero observations; otherwise the affected nonzero contribution is a warm-up missing value.

$$a_X(h) = sqrt\{\, mean[X_j^2 : j \text{ is retained and } time(j) < h] \,\}, \quad X \in \{H, B, H_k, B_k\}$$

$$S_i = 0.25 \times H_i / a_H(h) + 0.75 \times B_i / a_B(h)$$

3.2 Hourly aggregation and the 12 measures

Document scores are summed within UTC hour h. The same construction is repeated for each of the five categories. A second family divides each net index by its corresponding number of relevant documents, yielding average tone. The result is six net measures plus six average-tone measures.

$$I_h^{total} = Sum_{i \text{ in } U(h)}\, S_i; \quad Ibar_h^{total} = I_h^{total} / N_h^{relevant}$$

Family	Total	Loss of control	Malicious use	Job displacement	Discrimination	Privacy
Net	AI_Risk_total	AI_Risk_loss...	AI_Risk_malicious...	AI_Risk_job...	AI_Risk_discrimination	AI_Risk_privacy...
Average tone	total_average_tone	loss_average_tone	malicious_average_tone	job_average_tone	discrimination_average_tone	privacy_average_tone

The total index is not the sum of the five category indices. Categories overlap for multi-label units, and generic 0.2 risk contributes only to the total series. In 253 observed real hours, the direct total differs from the category sum.

3.3 Coverage and audit

Audit item	Result	Interpretation
Documents	1,882 raw; 1,859 retained	23 near-duplicate clusters removed
Relevant documents	447 raw; 440 retained	At least one relevant headline or clause
Hourly grid	723 hours	302 computed; 323 observed zero-risk; 89 unknown; 9 warm-up
Eligible hours	622 observed-sample; 0 strict	All keyword queries hit their result cap

Unknown empty hours are NaN, not zero. Accordingly, every empirical result below refers to an observed X News sample index, not a complete census of AI-risk news.

4 Index Dynamics and Financial Data

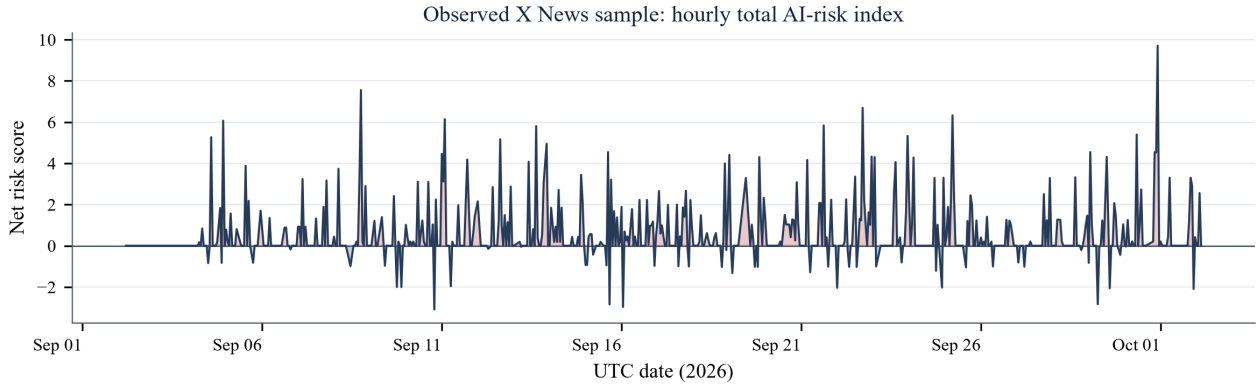


Figure 1. Hourly total AI-risk net score for the 622 observed-sample eligible hours. Positive values indicate net risk-affirming language; negative values indicate net denial or mitigation. Line gaps reflect unknown collection coverage or warm-up, not assumed zero risk.

The total index has mean 0.589, standard deviation 1.417, minimum -3.092, and maximum 9.700. It is nonzero in 279 of 622 observed hours. Positive bursts dominate because 67.6% of broadly relevant labeled units affirm or warn about risk. Category series are much sparser: the discrimination index is nonzero in only seven observed hours, while loss of control, malicious use, job displacement, and privacy are nonzero in 73, 70, 58, and 51 hours, respectively.

4.1 Intraday returns

The equity sample contains 20 firms with substantial AI exposure: MSFT, GOOGL, META, AMZN, TSLA, ORCL, CRM, NOW, ADBE, WDAY, INTU, PATH, AAPL, PLTR, UBER, SNOW, SHOP, RBLX, SNAP, and PINS. Yahoo consolidated five-minute raw/unadjusted OHLC bars are aggregated into six exact New York windows per trading day, from 10:00-11:00 through 15:00-16:00. The primary outcome is each window's open-to-close log return, avoiding any need to bridge the 09:30-10:00 half hour.

Dataset	Period / sessions	Coverage	Construction
20-stock panel	Sep 1-Oct 2; 23 NYSE days	138 hours/stock; 2,760 rows; complete	Raw 5-minute bars to exact 60-minute OHLC
QQQ	Same sessions	138/138 hours complete	Same source and aggregation
Premarket bridge	07:00-10:00 NY	887/1,380 stock-hours complete	Lag construction only; never a target; no hour covers all 20 stocks

No prices or returns are forward-filled, interpolated, or set to zero. Each regular-session target window passed duplicate-time, ordering, positive-price, OHLC, split-event, and completeness checks; incomplete premarket windows remain explicitly missing.

4.2 Strict temporal alignment

Risk measured in UTC hour [t-1, t) predicts the market return in [t, t+1). No news from the return window is included in its own predictor. Ticker fixed effects absorb persistent cross-firm differences, while target-hour and weekday fixed effects absorb recurring intraday and calendar patterns. This timing is designed to prevent look-ahead within the dataset; it cannot prove that every cluster update was instantly observable or eliminate all pretraining-related model knowledge.

5 Granger-Style Predictability Tests

5.1 Specification

The unrestricted model augments a return autoregression with one to three lags of the hourly total risk index. The return-lag order q is selected by Bayesian Information Criterion (BIC); risk-lag orders p = 1, 2, 3 are then compared on a common eligible sample. The joint null is that all coefficients on lagged risk equal zero. BIC rewards fit but penalizes added parameters, so it discourages mechanically selecting the longest lag structure.

$$r_{i,t} = alpha_i + FE_{hour(t)} + FE_{weekday(t)} + Sum_{j=1..q}\, phi_j\, r_{i,t-j} + Sum_{j=1..p}\, beta_j\, Risk_{t-j} + epsilon_{i,t}$$

For the 20-stock panel, standard errors are clustered two ways by ticker and target timestamp. The portfolio model requires a fixed 20-stock equal-weight return and reports day-clustered inference with HAC as a diagnostic. Benjamini-Hochberg false-discovery-rate adjustments are applied across the three lag specifications.

Return lags q	Panel BIC	Portfolio BIC
1	-14,910.522 (selected)	-601.231
2	-14,903.961	-597.130
3	-14,910.116	-603.642 (selected)

5.2 Main results

Model	Risk lags p	N	Cum. +1 SD effect	95% CI	Raw p	BH q
20-stock panel	1	1,789	+0.0177%	[-0.0301%, 0.0655%]	0.4479	0.8003
20-stock panel	2	1,789	+0.0236%	[-0.0406%, 0.0879%]	0.7210	0.8003
20-stock panel	3	1,789	+0.0116%	[-0.0818%, 0.1051%]	0.8003	0.8003
Equal-weight portfolio	1	54	+0.0325%	[-0.0516%, 0.1166%]	0.4264	0.8581
Equal-weight portfolio	2	54	+0.0175%	[-0.0673%, 0.1023%]	0.6969	0.8581
Equal-weight portfolio	3	54	+0.0307%	[-0.0929%, 0.1545%]	0.8581	0.8581

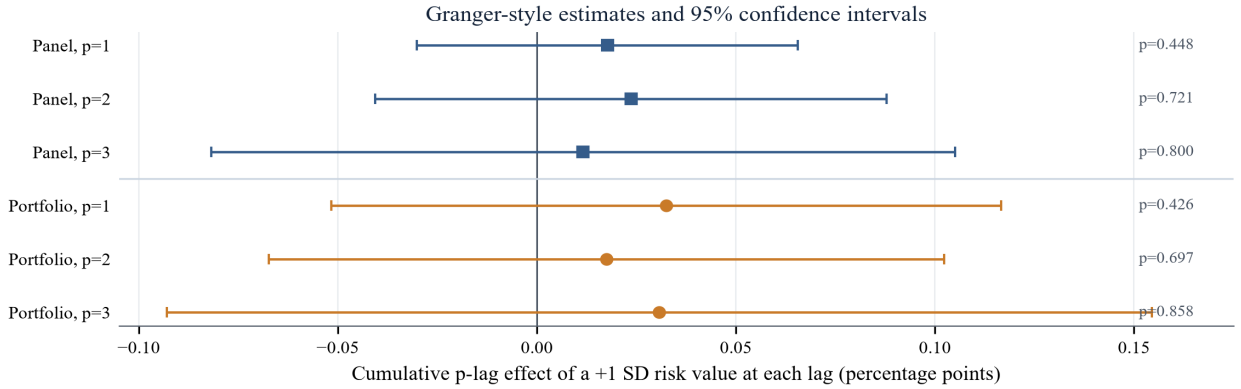


Figure 2. Each point is the cumulative p-lag effect: the sum of included lag coefficients multiplied by a one-SD risk value at every included lag (SD = 1.4168). Every 95% interval includes zero. The positive point estimates are small and run opposite to the original project hypothesis of a negative short-run effect.

6 Heterogeneity and Exploratory Results

6.1 Five risk categories

The equal-weight portfolio test is repeated separately for the five concrete risk categories. Of 15 planned category-by-lag specifications, ten are estimable and five lack within-sample lag variation. No raw p-value is below 0.05, and no Benjamini-Hochberg adjusted result is significant. The one-lag specifications below summarize the most directly comparable estimates.

Category (p=1)	N	Cum. +1 SD effect	95% CI	Raw p	BH q	Observed nonzero hours
Loss of human control	53	+0.0143%	[-0.0211%, 0.0497%]	0.4070	0.6784	73
Malicious use	54	+0.0289%	[-0.0368%, 0.0947%]	0.3665	0.6784	70
Job displacement	51	-0.0241%	[-0.0670%, 0.0189%]	0.2536	0.6784	58
Discrimination	Not estimable	-	-	-	-	7
Privacy violation	51	+0.0331%	[-0.0379%, 0.1041%]	0.3381	0.6784	51

Discrimination is not estimable at any lag order. Privacy is estimable only at p=1. This is a data-variation problem, not evidence that those risks have exactly zero market effect.

6.2 Individual stocks

Separate stock regressions generate 60 stock-by-risk-lag tests (20 stocks by three risk-lag choices), with each stock's return lag selected by BIC. Four specifications have an unadjusted joint-test $p<0.05$, but none survives either Benjamini-Hochberg or Holm correction. They should be treated as exploratory leads rather than discoveries.

Ticker	p	q	N	Cum. +1 SD effect	Raw p	BH q	HAC p
NOW	1	3	54	+0.1744%	0.0058	0.3100	0.0007
NOW	2	3	54	+0.1632%	0.0103	0.3100	0.0010
NOW	3	3	54	+0.2253%	0.0183	0.3528	0.0025
MSFT	1	1	95	+0.0616%	0.0235	0.3528	0.0035

The smallest adjusted q-value is 0.310. The HAC diagnostics are smaller for the same four rows, but they do not resolve the family-wise multiplicity problem, the short sample, or the fact that model selection and signal discovery use the same month of data.

6.3 What the null results mean

The current evidence says that lagged variation in this measured X-based hourly risk factor does not add stable one-to-three-hour predictive power beyond recent returns and calendar controls. It does not show that AI risk is irrelevant to valuation, that longer horizons are uninformative, or that a more complete and better-timestamped news measure would fail.

7 Trading-Strategy Diagnostic

A simple market-neutral-style diagnostic asks whether elevated risk helps a trader avoid AI-specific exposure. At each eligible market hour, the signal uses only the preceding news hour. High risk means the prior score is strictly above an expanding 75th percentile calculated from at least 72 still-earlier eligible risk hours. The strategy then holds 50% long QQQ and 50% short the fixed 20-stock equal-weight basket, closes at each day end, and pays 5 bp per one-way unit of drift-aware turnover.

Signal / portfolio item	Observed value
Market sample	138 hours across 23 trading days
Threshold available	108 hours; 100 signal-ready and 8 risk-missing
High-risk exposure	31 hours across 14 days; 21 entries
Portfolio	+50% QQQ / -50% fixed AI equal-weight; 100% gross, zero net
Turnover	42.04 one-way units including daily closure

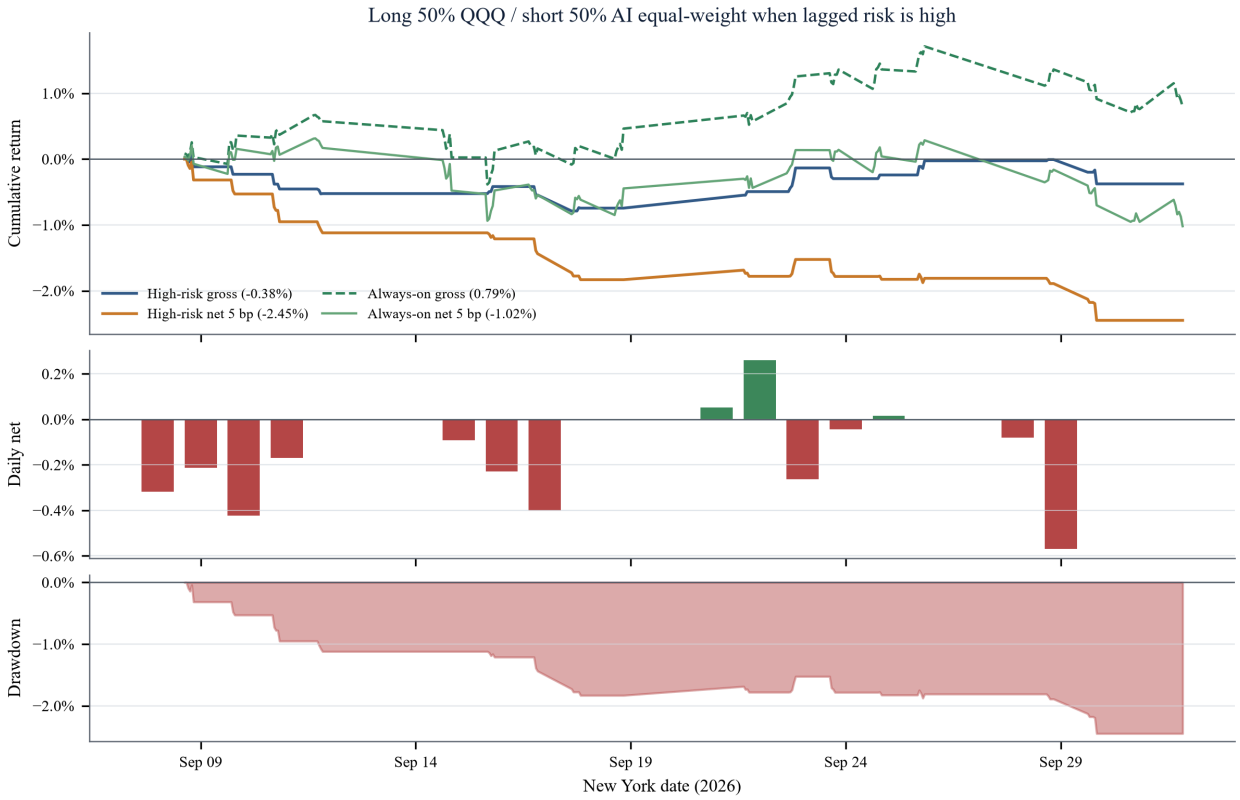


Figure 3. High-risk strategy cumulative gross and net returns, daily net returns, and net drawdown. The gross strategy ends at -0.38%; after the 5 bp turnover charge it ends at -2.45%, equal to its maximum drawdown. The always-on spread earns +0.79% gross but loses 1.02% net after its own costs.

7.1 Performance and sensitivity

Metric	High-risk strategy	Always-on spread
Gross cumulative return	-0.3766%	+0.7947%
Net cumulative return, 5 bp	-2.4497%	-1.0197%
Cumulative modeled cost	2.1020%	1.8158%
Win rate	51.61% of active hours	54.63% of hours
Annualized hourly Sharpe	-12.654	-2.358
Maximum drawdown	-2.4497%	-1.3303%

On high-risk hours, the QQQ leg contributes +0.1289%, while the AI short leg contributes -0.5041%, producing approximately -0.3752% gross. The active gross mean is -0.0121% per hour and is not significant when clustered by day (p=0.637). High-risk minus non-high-risk gross return is -0.0554 percentage points (p=0.114). The statistically negative mean after costs reflects the modeled transaction charge, not proof of negative gross alpha.

Sensitivity	Setting	Net cumulative return
Risk threshold	70% / 75% / 80% / 90%	-2.51% / -2.45% / -2.34% / -1.52%
One-way cost at 75%	0 / 2 / 5 / 10 bp	-0.38% / -1.21% / -2.45% / -4.48%

8 Robustness, Measurement Risk, and Limitations

- Incomplete news coverage. Every X query hit its result cap; strict coverage contains zero usable hours. Eighty-nine empty hours are correctly left unknown rather than converted to zero.
- Source semantics. X News records are evolving clusters with summaries and updated_at times, not full publisher stories with verified first-publication timestamps. Guardian full text has not yet entered the empirical pipeline.
- Annotation uncertainty. The pipeline has 794 review flags, including three nonzero clause scores. The 0.2 generic-risk rule is a design choice, not a confidence probability, and it contributes only to the total index.
- Small and sparse sample. The return sample spans 23 trading days. Category variation is thin, the portfolio Granger model has only 54 complete observations, and inference relies on few day clusters.
- Multiple testing. Four individual-stock signals are nominally significant, but all disappear after correction. Sparse unestimable category tests cannot be interpreted as zero effects.
- Trading frictions and exposure. The backtest omits processing latency, dynamic spreads, price impact, financing, and borrow fees. QQQ overlaps economically with the AI basket, so the spread is not a clean non-AI hedge.
- Interpretation. Granger-style rejection or non-rejection concerns incremental prediction in this sample. It does not identify a causal channel from news to prices.

9 Conclusion

This project turns a difficult theoretical construct into a reproducible hourly measurement pipeline. It distinguishes five concrete forms of AI-related harm from generic risk language, preserves evidence at the headline and clause level, prevents contemporaneous news from entering the predicted return, and separates exploratory signals from multiplicity-adjusted evidence. These design choices are a meaningful contribution even though the return tests are null.

Within the observed Sep 2-Oct 2, 2026 X News sample, the total risk index does not robustly predict next-hour AI-stock returns at one to three lags. Category tests are underpowered and nonsignificant; stock-specific nominal findings do not survive correction; and the proposed high-risk long-QQQ/short-AI trade is negative before costs and substantially worse after costs. The appropriate conclusion is therefore narrow: the current AI-risk factor and trading rule are not validated as short-horizon return predictors. A longer, uncapped, publisher-timestamped sample with completed Guardian labeling and true out-of-sample testing is needed before judging the construct more generally.

The original second hypothesis - that AI-development news positively predicts AI-company returns - remains open because no development taxonomy, labels, index, or regression has yet been implemented.

Data and Code Availability

The project workspace contains the collection, preprocessing, DeepSeek annotation, index-construction, hourly-return, Granger-regression, and backtest programs; source and processed data reside under data/. Primary analysis notebooks are stored at the repository root. API credentials are read from the environment and are not included in this report.

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Appendix A. Descriptive Statistics for the 12 Hourly Measures

Measure	n	Nonzero	Mean	SD	Min	Max
Total net	622	279	0.589	1.417	-3.092	9.700
Loss-control net	618	73	0.228	0.942	-4.575	8.559
Malicious-use net	616	70	0.393	1.508	-2.181	11.958
Job-displacement net	614	58	0.304	1.540	-7.119	14.267
Discrimination net	617	7	0.032	1.120	-16.822	14.671
Privacy net	615	51	0.051	1.400	-6.366	19.317
Total average tone	301	279	0.879	1.371	-2.098	6.323
Loss-control avg.	77	73	1.725	1.915	-4.575	6.771
Malicious-use avg.	71	70	3.279	3.021	-2.181	11.958
Job-displacement avg.	63	58	2.749	3.609	-7.119	13.238
Discrimination avg.	8	7	2.465	10.177	-16.822	14.671
Privacy avg.	55	51	0.545	4.617	-6.366	19.317

Notes: Net statistics use the 622 observed-sample eligible hours where available. Average tone is defined only when at least one corresponding relevant document appears in the hour, so its missingness is intentional. Extreme category averages can arise when sparse, scaled category scores are divided by one relevant document.